

Saurav Kumar

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

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EDUCATION

- **Technische Universität Darmstadt** Darmstadt, Germany
Semester Exchange Student; CGPA: 8.44
Oct 2025 – March 2026
 - **Relevant Coursework:** Artificial Intelligence, Computer Vision, Probabilistic Graphical Models, LLMs, Scientific Computing
- **Indian Institute of Technology, Mandi** Mandi, Himachal Pradesh, India
B.Tech in Bioengineering; CGPA: 8.02/10
Aug 2023 – Present
 - **Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Python & Data Science, Probability & Statistics

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, Perl | **Frameworks:** React, ROS, TensorFlow, Django, Streamlit
Tools: Git, Linux, Docker, SolidWorks, Raspberry Pi, Arduino | **Concepts:** Sensor Fusion, Kalman Filtering, Control Systems

EXPERIENCE

- **Morsebiz** Remote
Web Development Intern
Jan 2026 – Mar 2026
 - **AI Data Processing:** Developed high-throughput web scraping pipelines coupled with an AI-driven content verification system, ensuring the extraction and sanitization of unstructured data with high accuracy.
 - **System Integration:** Engineered automated communication workflows by integrating asynchronous messaging APIs, enabling real-time system alerts and seamless cross-platform user engagement.
 - **Automated Billing Ops:** Architected a dynamic, automated invoice generation module that streamlined billing operations, significantly reducing manual overhead and ensuring strict transactional compliance.

PROJECTS

- **Predicting Protein Stability Changes ($\Delta\Delta G$)** [Live](#) | [Paper](#)
Machine Learning, Siamese GNN, ESM-2, Bayesian Networks
Feb 2026 – Present
 - **Architecture Design:** Engineered a 1D-3D latent fusion architecture combining a Siamese Graph Neural Network (GNN) with a 150-million parameter Protein Language Model (ESM-2).
 - **Predictive Performance:** Achieved a Pearson Correlation of 0.7266 and Mean Absolute Error of 0.7472 kcal/mol, significantly surpassing traditional biophysical force fields.
 - **Causal Inference:** Constructed a Hybrid Bayesian Network to map explicit biological causality, mathematically quantifying the vulnerability of protein surfaces to polarity shifts.
- **Vaya – Your Local Healthcare Connection** [GitHub](#)
Python, Django, Django Channels, Groq API, RAG
Aug 2025 – Present
 - **Full-stack Platform:** Designed doctor discovery, appointment booking, and report handling system with secure authentication.
 - **AI RAG System:** Implemented a two-step retrieval-augmented generation pipeline reducing hallucinations and improving accuracy.
 - **Real-time Messaging:** Built encrypted, real-time patient-doctor chat with Django Channels.
- **AquaSweep – Underwater Rover** [GitHub](#)
Python, OpenCV, Flask, ROS, Arduino, Raspberry Pi
Jan 2025 – Jun 2025
 - **Live Streaming:** Engineered MJPEG camera pipeline (20 FPS, ≈ 200 ms latency) using Flask sockets.
 - **Edge AI Detection:** Built OpenCV pipelines for algae and motion detection with 93% lab accuracy.
 - **Depth Stabilization:** Controlled 8 thrusters via ROS-Arduino; fused IMU and Bar30 data with Kalman filtering (± 3 cm error).

LEADERSHIP & ACHIEVEMENTS

Awards & Honors: Selected for TU Darmstadt semester exchange (Top 450+ batch merit) | Secured Top 5 rank in Arduino Hackathon.
Extracurriculars: Hospitality Head for Exodia (managed 1000+ participants) | Robotics Club Mentor, IIT Mandi.